

Chapter 5: Mathematics Logic

Class 22: Practical / Activities (20 Minutes)

Topic: Mental Mathematics Race, Sudoku Practice, Strategy Game Challenge & Team Problem-Solving Activity

Activity 1: Mental Mathematics Race (5 Minutes)

Objective

To improve students speed, accuracy, concentration, and confidence in performing mental calculations without using paper or calculators.

Materials Required

- Number Flashcards
- Whiteboard & Marker
- Timer or Stopwatch
- Scoreboard
- Student Response Cards (Optional)

Instructions

1. Divide the class into two or more teams.
2. The teacher displays one mental mathematics question at a time.
3. Students solve the question mentally and raise their hands to answer.
4. The first correct answer earns **1 point** for the team.
5. Continue with different types of questions and announce the winning team.

Mental Mathematics Questions

Addition

- $25 + 35 =$ _____
- $48 + 22 =$ _____
- $120 + 80 =$ _____

Subtraction

- $90 - 35 =$ _____
- $150 - 75 =$ _____
- $200 - 125 =$ _____

Multiplication

- $8 \times 7 =$ _____
- $12 \times 5 =$ _____
- $9 \times 9 =$ _____

Division

- $72 \div 8 =$ _____
- $100 \div 10 =$ _____
- $81 \div 9 =$ _____

Teacher Observation Checklist

- ☒ Solves problems mentally.
 - ☒ Answers quickly and accurately.
 - ☒ Demonstrates confidence.
 - ☒ Participates actively.
 - ☒ Shows improvement in calculation speed.
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Activity 2: Sudoku Practice (5 Minutes)

Objective

To develop logical thinking, observation, concentration, and reasoning skills through a simple Sudoku puzzle.

Materials Required

- 4×4 Sudoku Worksheet
- Pencil & Eraser
- Whiteboard

- Teacher Answer Key

Instructions

1. Explain the basic Sudoku rules.
2. Distribute the Sudoku worksheet to all students.
3. Students complete the puzzle independently.
4. The teacher discusses the correct solution after completion.
5. Encourage students to explain their reasoning.

Sudoku Rules

- Use the numbers **1, 2, 3, and 4**.
- Each number must appear **only once** in every row.
- Each number must appear **only once** in every column.
- Do not repeat numbers in the same row or column.

Practice Puzzle

1	3
3	1
2	4
4	2

Teacher Observation Checklist

- ☒ Understands Sudoku rules.
 - ☒ Uses logical reasoning.
 - ☒ Completes the puzzle correctly.
 - ☒ Works independently.
 - ☒ Demonstrates concentration.
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Activity 3: Strategy Game Challenge (5 Minutes)

Objective

To strengthen students' planning, logical thinking, teamwork, and mathematical reasoning through fun strategy games.

Materials Required

- Number Cards
- Dice
- Puzzle Cards
- Domino Cards (Optional)
- Whiteboard
- Timer

Instructions

1. Divide students into small groups.
2. Assign one strategy game to each group.
3. Students work together to complete the challenge.
4. The group that finishes correctly within the time limit earns points.
5. Rotate games if time permits.

Strategy Game Options

Game 1 – Number Relay

Students arrange number cards in the correct order from smallest to largest.

Game 2 – Number Bingo

The teacher calls out answers to simple calculations, and students mark the corresponding numbers on their Bingo cards.

Game 3 – Puzzle Race

Students solve simple number puzzles and pattern challenges within the time limit.

Game 4 – Domino Mathematics

Students match domino cards by solving addition or multiplication problems.

Teacher Observation Checklist

- ☒ Works effectively in a team.
 - ☒ Uses logical thinking.
 - ☒ Completes tasks accurately.
 - ☒ Demonstrates good planning.
 - ☒ Participates enthusiastically.
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Activity 4: Team Problem-Solving Activity (5 Minutes)

Objective

To develop teamwork, communication, and mathematical problem-solving skills by solving real-life situations collaboratively.

Materials Required

- Problem-Solving Worksheet
- Whiteboard
- Pencil
- Chart Paper (Optional)

Instructions

1. Divide students into groups of four or five.
2. Give each group a worksheet containing mathematical word problems.
3. Students discuss and solve each problem together.
4. One member presents the group's solution to the class.
5. The teacher provides feedback and discusses different solution methods.

Problem-Solving Questions

Problem 1

A library has **120 books**. It buys **35 more books**.

How many books are there now?

Answer: 155 books

Problem 2

A school bus has **50 seats**. **42 students** are sitting.

How many seats are empty?

Answer: 8 seats

Problem 3

A fruit shop has **85 apples**. It sells **27 apples**.

How many apples are left?

Answer: 58 apples

Problem 4

A student reads **15 pages** every day.

How many pages will the student read in **4 days**?

Answer: 60 pages

Teacher Observation Checklist

☒ Understands the problem.

- ✓ Applies correct mathematical operations.
 - ✓ Explains the solution clearly.
 - ✓ Works cooperatively with teammates.
 - ✓ Participates confidently.
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Practical Time Distribution

Activity	Duration
Mental Mathematics Race	5 Minutes
Sudoku Practice	5 Minutes
Strategy Game Challenge	5 Minutes
Team Problem-Solving Activity	5 Minutes
Total Practical Time	20 Minutes

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Perform **mental calculations** quickly and accurately without using calculators.
- Solve **Sudoku puzzles** by applying logical reasoning and observation skills.
- Participate effectively in **strategy games** that enhance planning, teamwork, and mathematical thinking.
- Solve **real-life mathematical problems** using appropriate problem-solving strategies.
- Demonstrate improved **critical thinking, decision-making, communication, and collaboration** during group activities.
- Build confidence in applying **mental mathematics and logical reasoning** to classroom tasks and everyday situations.